



Road Safety Data Analysis Dashboard Report

CiSTUP Internship Application - Round 1

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Table Of Contents

Content	Page Number
Executive Summary	3
Dashboard Architecture & Design Approach	3
Data Preparation & Quality Assessment	4
Additional Measures & DAX measures for Dashboard creation	9
Key Performance Indicators (KPIs) Analysis	12
Visuals & Dashboard Creation in Power BI	15
Technical Implementation & Dashboard Features	19
Methodology & Analytical Approach	19
Strategic Recommendations for Infrastructural Planning	20
Conclusion	21

Executive Summary

This report presents a comprehensive Power BI dashboard analyzing road safety data to support infrastructure planning and policy decisions. The dashboard provides multi-dimensional insights into accident patterns, environmental factors, temporal trends, and geographic hotspots through interactive visualizations.

Key Findings

- **308,000 total accidents** analyzed, with **68.16% data quality**, ensuring reliable insights
- **3,953 fatal accidents** represent the most critical safety concern
- **418,000 total casualties**, indicating a significant socioeconomic impact
- Clear temporal patterns with **peak accident hours** between 15:00hrs and 18:00hrs.
- **Urban areas account for 64.47%** of all accidents, highlighting infrastructure challenges

Dashboard Architecture & Design Approach

Multi-Page Structure

The dashboard is organized into five analytical layers:

1. **Executive Summary Dashboard** - High-level KPIs and overview metrics
2. **Temporal Analysis** - Time-based patterns and peak hour analysis
3. **Environmental & Infrastructure Analysis** - Weather and lighting conditions impact
4. **Geographic Analysis** - Spatial distribution and risk hotspots
5. **Vehicle Types & Casualty Analysis** - Vehicle-specific insights

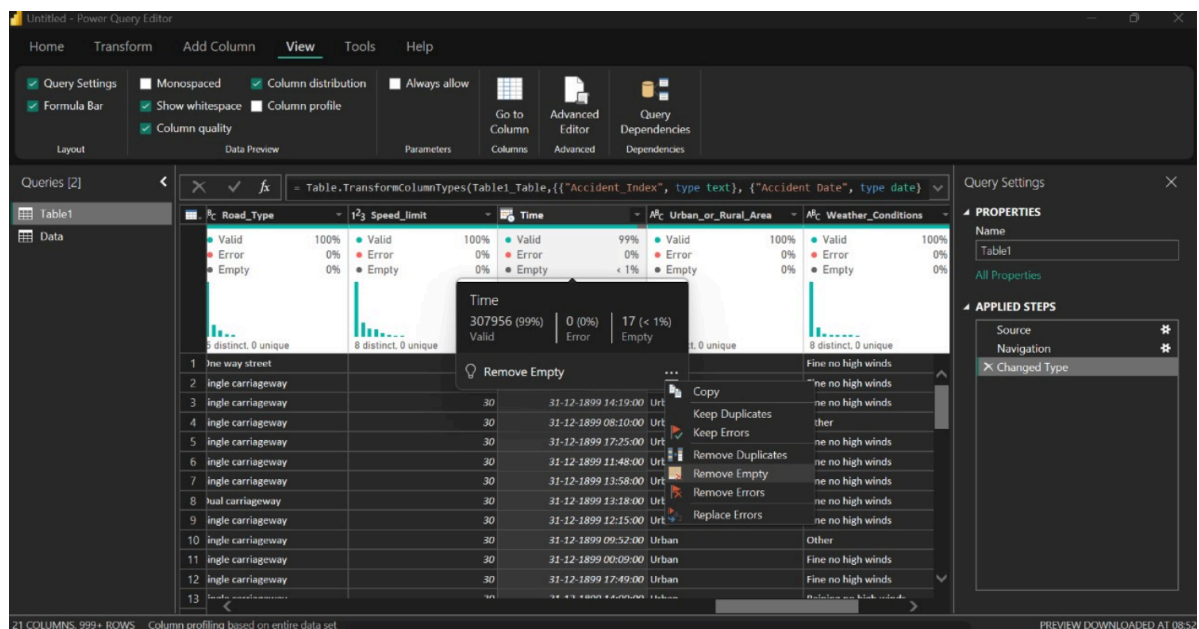
Design Principles

- **User-centric navigation** with consistent filter panels
- **Interactive cross-filtering** enabling drill-down analysis
- **Professional branding** with IISc institutional identity
- **Color-coded severity levels** for immediate visual impact
- **Responsive layout** optimized for different screen sizes

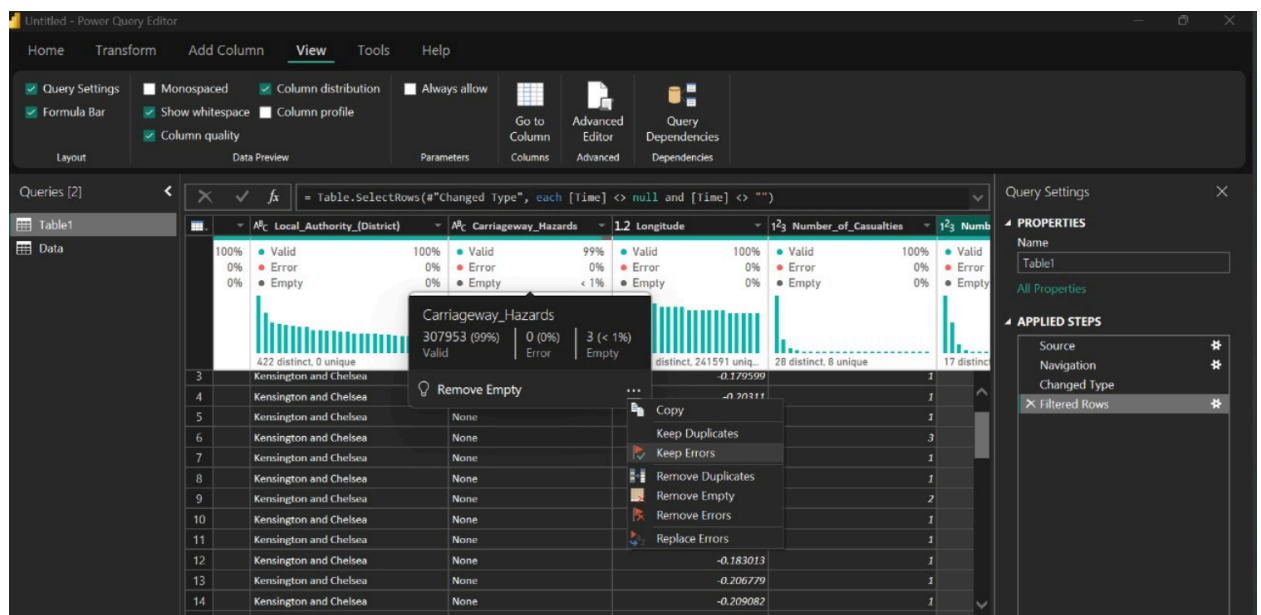
Data Preparation & Quality Assessment

Data Cleaning Process

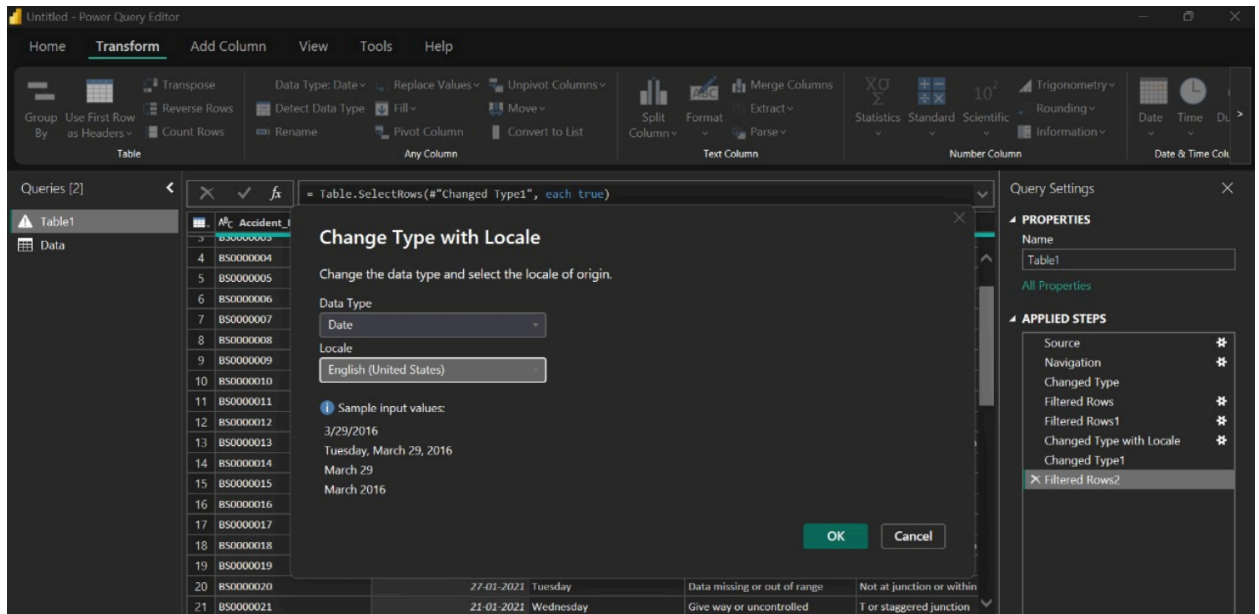
1. **Handling Empty time values (17)** - Deleting the empty time rows, as time is very crucial in any road accident, and we cannot predict the missing value.



2. **Removing empty values from Carriage_Hazards (3)**



3. Formatting all the date values to a compatible date type



4. Changing the Time column to only required time, and deleting the date involved for better understanding

ansformColumnTypes("#Changed Type with Locale1",{{"Time"

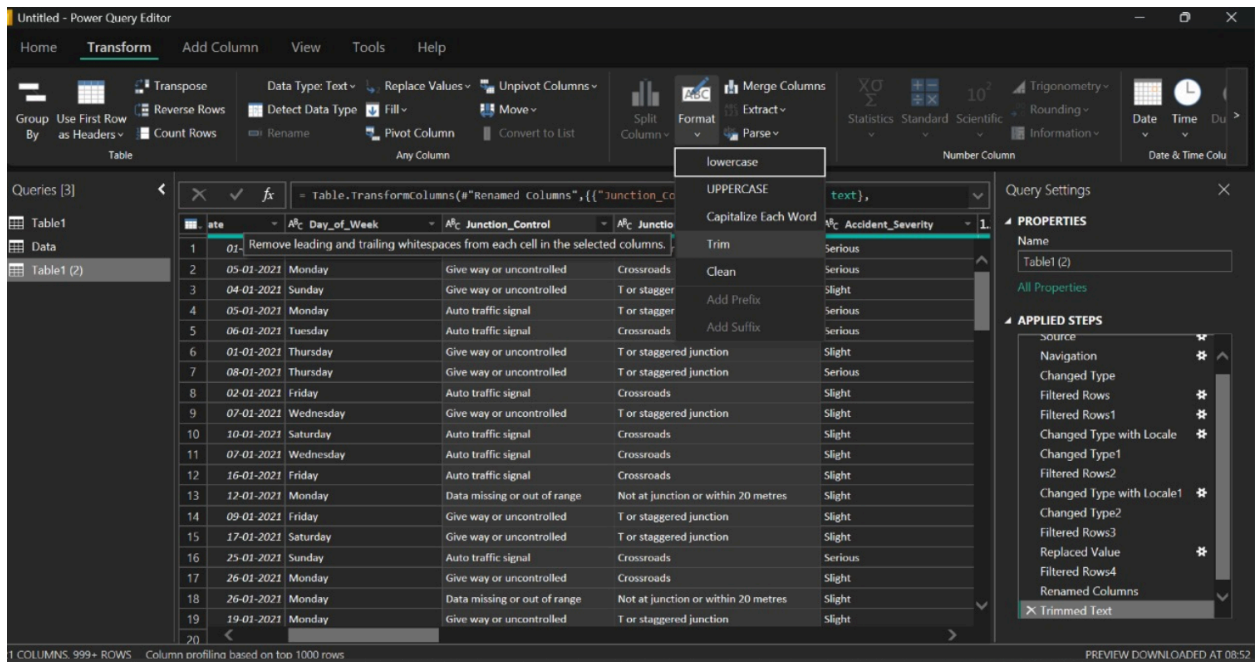
	Time	Urban_or_Rural_Area
30	15:11:00	Urban
30	10:59:00	Urban
30	14:19:00	Urban
30	08:10:00	Urban
30	17:25:00	Urban
30	11:48:00	Urban
30	13:58:00	Urban
30	13:18:00	Urban
30	12:15:00	Urban
30	09:52:00	Urban
30	00:09:00	Urban
30	17:49:00	Urban
30	14:00:00	Urban
30	08:15:00	Urban
30	12:15:00	Urban
30	22:05:00	Urban
30	17:30:00	Urban
30	17:05:00	Urban
30	14:27:00	Urban

5. Renaming the Accidental Severity column data values - Fetal -> Fatal.

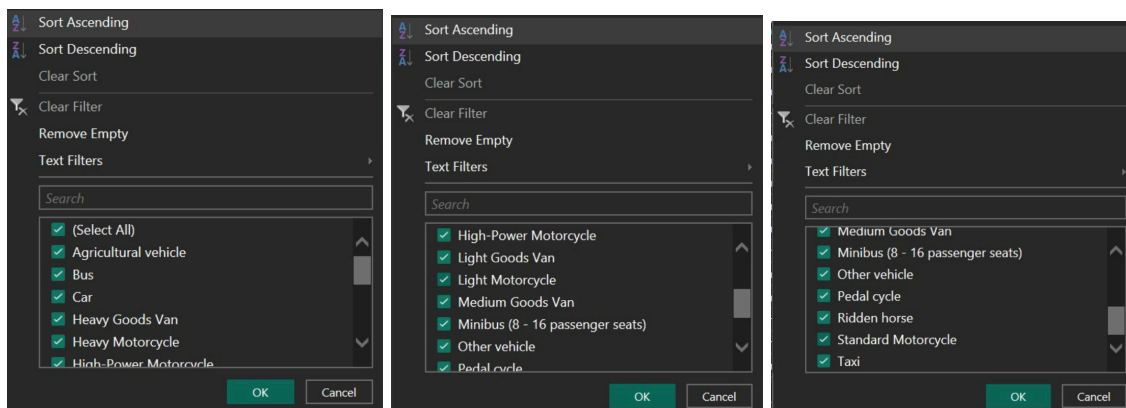
The screenshot shows the Power Query Editor interface. The 'Transform' tab is active. A context menu is open over the 'Accident_Severity' column, showing options like 'Sort Ascending', 'Sort Descending', 'Clear Sort', 'Clear Filter', 'Remove Empty', and 'Text Filters'. The 'Text Filters' option is selected, and a sub-menu is displayed with checkboxes for '(Select All)', 'Fatal', 'Fetal', 'Serious', and 'Slight'. The 'Fatal' checkbox is checked. The main data table is visible in the background, showing columns for 'Junction_Detail', 'Latitude', 'Light_Conditions', and 'Location'. The 'Accident_Severity' column contains values like 'Fetal', 'Serious', and 'Slight'.

The screenshot shows the Power Query Editor interface. The 'Transform' tab is active. A context menu is open over the 'Accident_Severity' column, showing options like 'Sort Ascending', 'Sort Descending', 'Clear Sort', 'Clear Filter', 'Remove Empty', and 'Text Filters'. The 'Text Filters' option is selected, and a sub-menu is displayed with checkboxes for '(Select All)', 'Fatal', 'Fetal', 'Serious', and 'Slight'. The 'Fatal' checkbox is checked. The main data table is visible in the background, showing columns for 'Junction_Detail', 'Latitude', 'Light_Conditions', and 'Location'. The 'Accident_Severity' column contains values like 'Fetal', 'Serious', and 'Slight'.

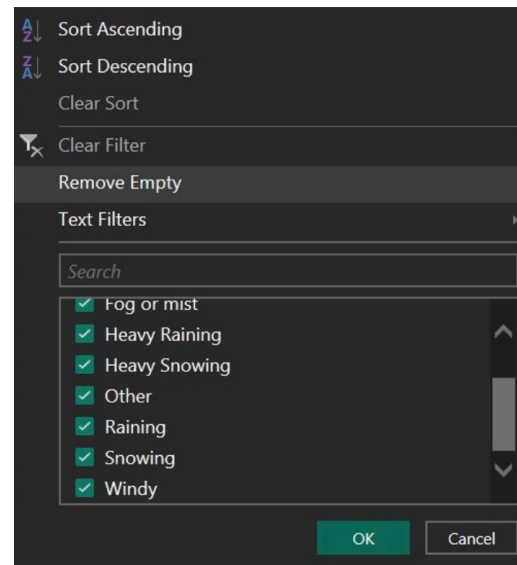
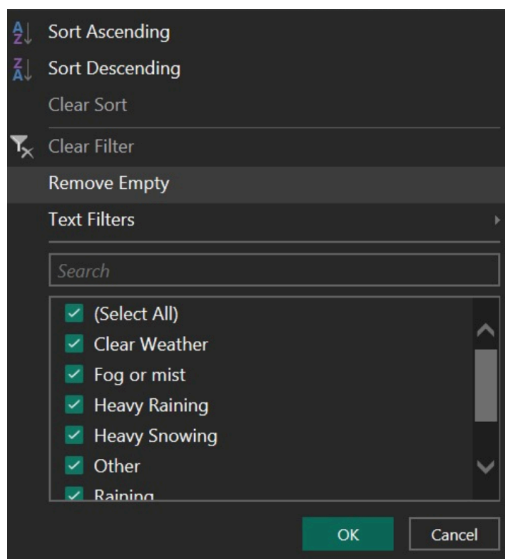
6. Performed trim operations to remove unwanted white spaces from all the columns.



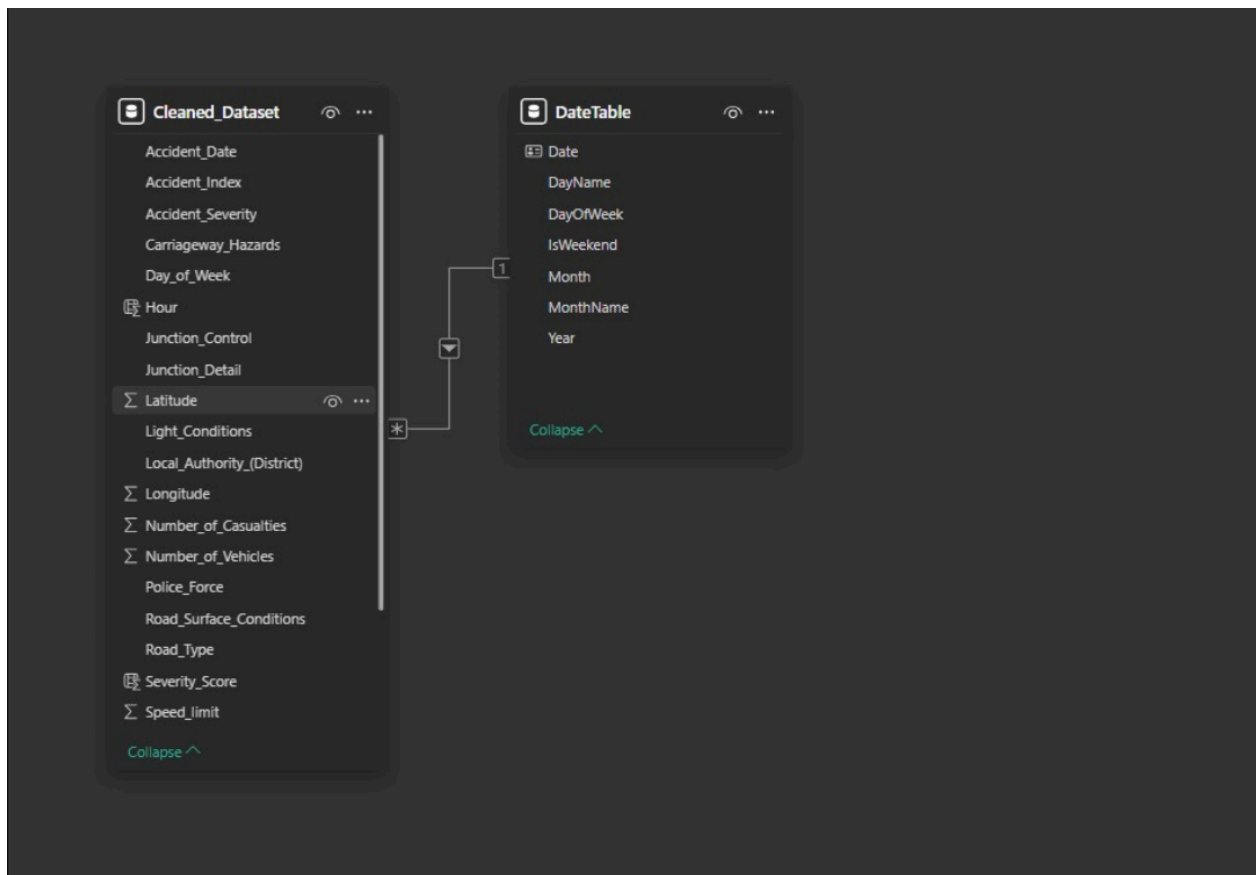
7. Simplified the Vehicle_Type column for more understandable and easier interpretation.



8. Simplified the Weather_Type column for a more understandable and easier interpretation.



9. Relationship Establishment for dashboard creation - created another table.



Additional Measures & DAX measures for Dashboard creation

1. Data Quality & Total Accidents -

```
1 DataQuality = (1 - DIVIDE(COUNTROWS(FILTER('Cleaned_Dataset', 'Cleaned_Dataset'[Junction_Control] = "Data  
missing or out of range")), [Total Accidents])) * 100
```

```
1 Total Accidents = COUNTROWS('Cleaned_Dataset')
```

2. Day & Hour-

```
1 Day = DAY('Cleaned_Dataset'[Accident_Date])
```

```
1 Hour = HOUR('Cleaned_Dataset'[Time])
```

3. Fatal Accidents -

```
1 Fatal Accidents = CALCULATE(COUNTROWS('Cleaned_Dataset'), 'Cleaned_Dataset'[Accident_Severity] = "Fatal")
```

4. Junction Risk Score & Location Risk Score-

```
1 Junction Risk Score =  
2 CALCULATE(  
3     AVERAGE('Cleaned_Dataset'[Severity_Score]),  
4     ALLEXCEPT('Cleaned_Dataset', 'Cleaned_Dataset'[Junction_Control])  
5 )
```

```
1 Location Risk Score =  
2 CALCULATE(  
3     AVERAGE('Cleaned_Dataset'[Severity_Score]) * COUNT('Cleaned_Dataset'[Accident_Index]),  
4     ALLEXCEPT('Cleaned_Dataset', 'Cleaned_Dataset'[Latitude], 'Cleaned_Dataset'[Longitude])  
5 )
```

5. Month Name & Month Number & Year-

```
1 Month Name = FORMAT('Cleaned_Dataset'[Accident_Date], "MMM")
```

```
1 Month Number = MONTH('Cleaned_Dataset'[Accident_Date])
```

```
1 Year = YEAR('Cleaned_Dataset'[Accident_Date])
```

6. Monthly Accidents -

```
1 Monthly Accidents = CALCULATE([Total Accidents], DATESMTD(DateTable[Date]))
```

7. Peak Day & Peak Hour-

```
1 PEAK_DAY =
2 VAR MaxAccidents = MAXX(ALL(DateTable[DayName]), [Total Accidents])
3 RETURN
4 CALCULATE(
5     MAX(DateTable[DayName]),
6     FILTER(ALL(DateTable[DayName]), [Total Accidents] = MaxAccidents)
7 )
```

```
1 Peak_Hour =
2 VAR MaxAccidents = MAXX(ALL('Cleaned_Dataset'[Hour]), [Total Accidents])
3 RETURN
4 CALCULATE(
5     MAX('Cleaned_Dataset'[Hour]),
6     FILTER(ALL('Cleaned_Dataset'[Hour]), [Total Accidents] = MaxAccidents)
7 )
```

8. Risk Score -

```
1 Risk_Score =
2 DIVIDE(
3     CALCULATE(COUNT(Cleaned_Dataset[Accident_Index]), Cleaned_Dataset[Accident_Severity] = "Fatal") * 3 +
4     CALCULATE(COUNT(Cleaned_Dataset[Accident_Index]), Cleaned_Dataset[Accident_Severity] = "Serious") * 2 +
5     CALCULATE(COUNT(Cleaned_Dataset[Accident_Index]), Cleaned_Dataset[Accident_Severity] = "Slight") * 1,
6     COUNT(Cleaned_Dataset[Accident_Index])
7 )
```

9. Road Surface Risk -

```
1 Road Surface Risk =
2 CALCULATE(
3     AVERAGE('Cleaned_Dataset'[Severity_Score]),
4     ALLEXCEPT('Cleaned_Dataset', 'Cleaned_Dataset'[Road_Surface_Conditions])
5 )
```

10. Serious Accidents, Slight Accidents -

```
1 Serious Accidents = CALCULATE(COUNTROWS('Cleaned_Dataset'), 'Cleaned_Dataset'[Accident_Severity] = "Serious")
```

```
1 Slight Accidents = CALCULATE(COUNTROWS('Cleaned_Dataset'), 'Cleaned_Dataset'[Accident_Severity] = "Slight")
```

11. Severity Score & Weather Severity Index-

```
1 Severity_Score =  
2 SWITCH(  
3     Cleaned_Dataset[Accident_Severity],  
4     "Slight", 1,  
5     "Serious", 3,  
6     "Fatal", 5,  
7     0  
8 )
```

```
1 Weather Severity Index =  
2 CALCULATE(  
3     AVERAGE('Cleaned_Dataset'[Severity_Score]),  
4     ALLEXCEPT('Cleaned_Dataset', 'Cleaned_Dataset'[Weather_Conditions])  
5 )
```

12. Time Period -

```
1 Time_Period =  
2 SWITCH(  
3     TRUE(),  
4     Cleaned_Dataset[Hour] >= 6 && Cleaned_Dataset[Hour] < 12, "Morning (6AM-12PM)",  
5     Cleaned_Dataset[Hour] >= 12 && Cleaned_Dataset[Hour] < 18, "Afternoon (12PM-6PM)",  
6     Cleaned_Dataset[Hour] >= 18 && Cleaned_Dataset[Hour] < 22, "Evening (6PM-10PM)",  
7     "Night (10PM-6AM)"  
8 )
```

13. Weekday Accidents -

```
1 Weekday Accidents = CALCULATE([Total Accidents], DataTable[IsWeekend] = "Weekday")
```

14. Creation of a new table - DataTable for easier interpretation

```
1 DataTable =  
2 ADDCOLUMNS(  
3     CALENDAR(MIN('Cleaned_Dataset'[Accident_Date]), MAX('Cleaned_Dataset'[Accident_Date])),  
4     "Year", YEAR([Date]),  
5     "Month", MONTH([Date]),  
6     "MonthName", FORMAT([Date], "MMMM"),  
7     "DayOfWeek", WEEKDAY([Date], 2),  
8     "DayName", FORMAT([Date], "DDDD"),  
9     "IsWeekend", IF(WEEKDAY([Date], 2) >= 6, "Weekend", "Weekday")  
10 )
```

15. Day Order -

```
1 Day_Order =  
2 SWITCH(  
3     WEEKDAY(MAX('DataTable'[Date])),  
4     2, 1, -- Monday  
5     3, 2, -- Tuesday  
6     4, 3, -- Wednesday  
7     5, 4, -- Thursday  
8     6, 5, -- Friday  
9     7, 6, -- Saturday  
10    1, 7, -- Sunday  
11 )
```


Key Performance Indicators (KPIs) Analysis

1. Accident Severity Distribution

- **Fatal:** 41K accidents (13.23%)
- **Serious:** Significant portion requiring immediate attention
- **Slight:** 263K accidents (85.4%) - highest volume category

Insight: While slight accidents dominate in volume, fatal accidents represent critical infrastructure failure points requiring targeted intervention.

2. Temporal Patterns Analysis

Daily Accident Trends

- **Peak hours:** 15:00-18:00 showing maximum accident concentration
- **Morning rush:** Secondary peak during 07:00-09:00
- **Night hours:** Significantly lower but higher severity rates

Weekly Distribution

- **Weekdays:** Higher accident frequency (70.1%) correlating with commuting patterns
- **Weekends:** 29.9% of accidents but potentially higher severity rates
- **Friday:** Peak day with maximum accident occurrence

Insight: Accident patterns directly correlate with traffic volume and commuting behaviors, indicating infrastructure stress during peak hours.

3. Environmental Impact Analysis

Weather Conditions Matrix

- **Clear weather:** Highest absolute numbers (207,563 slight accidents)
- **Adverse conditions:** Higher severity rates despite lower volumes
- **Rain correlation:** Significant impact on accident severity distribution

Lighting Conditions

- **Daylight:** 60.90% of accidents (227.27K)
- **Darkness with lighting:** 19.51% showing infrastructure adequacy issues
- **Unlit areas:** Critical safety concern requiring immediate attention

Insight: Infrastructure lighting and road surface maintenance are critical factors in accident prevention, particularly during adverse weather conditions.

4. Geographic Risk Assessment

Urban vs Rural Distribution

- **Urban areas:** 64.47% of accidents reflecting population density and traffic complexity
- **Rural areas:** 35.53% but potentially higher fatality rates per incident

Speed Limit Analysis

- **30 mph zones:** Highest accident frequency but lower severity
- **Higher speed limits:** Fewer accidents but significantly higher severity rates

Insight: Speed management and urban traffic flow optimization are key infrastructure planning priorities.

5. Junction Control Effectiveness

Traffic Management Systems

- **Give way/uncontrolled:** 150K accidents - highest risk category
- **Data missing/out of range:** 98K incidents indicating data collection gaps
- **Auto traffic signals:** 32K accidents showing effectiveness of automated systems
- **Stop signs:** 2K accidents - most effective control method

Insight: Automated traffic management systems and proper signaling significantly reduce accident rates at junctions.

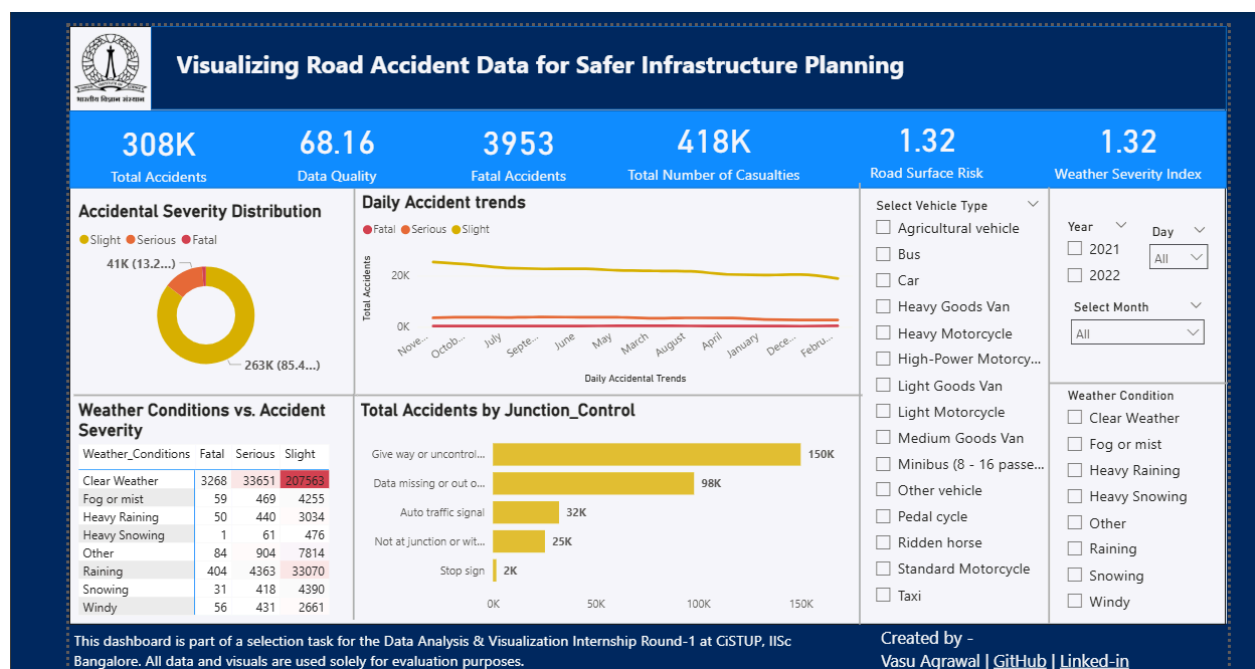
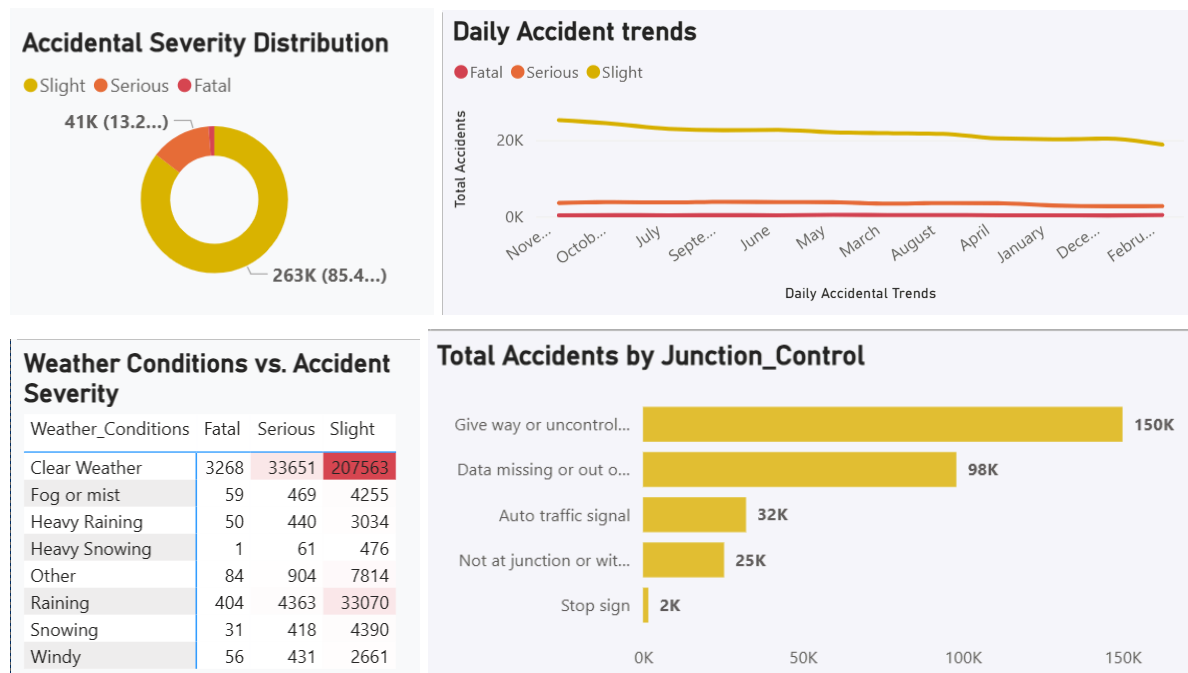
6. Vehicle Type Risk Profiling

High-Risk Categories

- **Cars:** Dominant in absolute numbers requiring targeted safety measures
- **Motorcycles:** High casualty rates per incident
- **Heavy goods vehicles:** Infrastructure damage and severity concerns
- **Vulnerable road users:** Cyclists and pedestrians requiring special protection

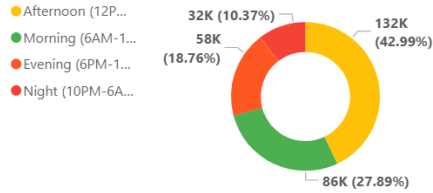
Visuals & Dashboard Creation in Power BI

Executive Summary Dashboard

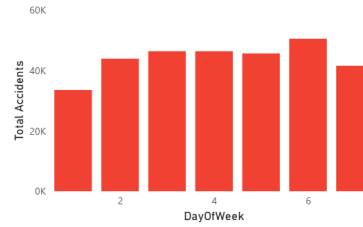


Temporal Analysis

Total Accidents by Time_Period



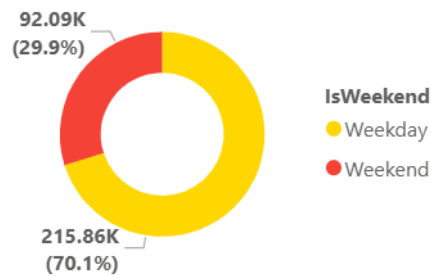
Accidents by Day Of Week



Peak Hour HeatMap Analysis

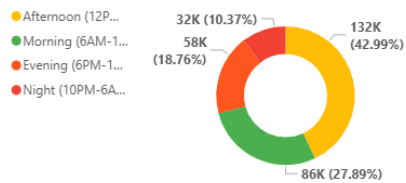
Hour	1	2	3	4	5	6	7
0	1187	482	380	403	497	558	1160
1	1054	340	258	251	310	360	960
2	806	213	165	171	207	325	765
3	760	183	146	153	205	258	606
4	504	147	141	152	185	199	393
5	377	337	318	346	336	359	418
6	418	855	884	894	872	822	529
7	571	2269	2338	2399	2175	1959	849
8	756	3942	4394	4426	4030	3712	1292
9	1066	2605	2680	2602	2514	2373	1720
10	1599	1960	2058	1984	1936	2096	2158
11	2057	2241	2259	2185	2131	2418	2850
12	2500	2483	2432	2477	2428	2843	3179
13	2514	2532	2569	2593	2584	3114	3065
14	2441	2584	2672	2586	2575	3261	2948
15	2368	3717	3691	3705	3608	4398	2663
16	2335	3780	3983	3890	3913	4291	2711
17	2263	4000	4488	4473	4509	4418	2813
18	2009	2948	3410	3458	3281	3476	2481
19	1695	1974	2356	2334	2424	2830	2238

Count of Accident_Index by IsWeekend

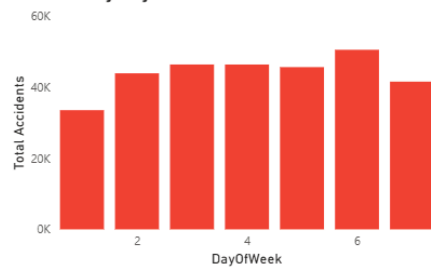


Temporal Analysis - Peak Hours & Patterns

Total Accidents by Time_Period



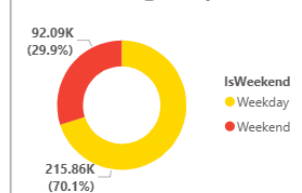
Accidents by Day Of Week



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Count of Accident_Index by IsWeekend



Year
☐ 2021
☐ 2022

Select Month

Day

Weather Condition
☐ Clear Weather
☐ Fog or mist
☐ Heavy Raining
☐ Heavy Snowing
☐ Other
☐ Raining
☐ Snowing
☐ Windy

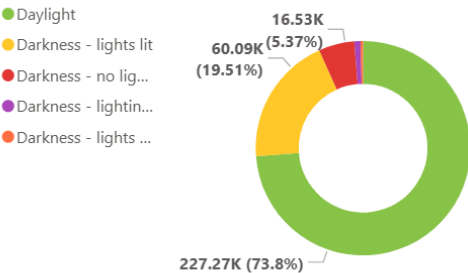
Select Vehicle
☐ Agricultural v...
☐ Bus
☐ Car
☐ Heavy Goods...
☐ Heavy Motor...
☐ High-Power ...

Environmental & Infrastructure Analysis

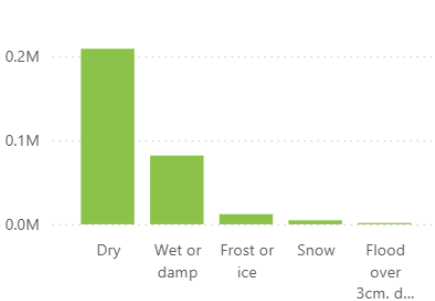
Weather Conditions Matrix

Weather_Conditions	Fatal	Serious	Slight
Clear Weather	3268	33651	207563
Fog or mist	59	469	4255
Heavy Raining	50	440	3034
Heavy Snowing	1	61	476
Other	84	904	7814
Raining	404	4363	33070
Snowing	31	418	4390
Windy	56	431	2661

Count of Accident_Index by Light_Conditions



Count of Accident_Severity by Road_Surface_Conditions



Weather Conditions Matrix by Lighting conditions

Weather_Conditions	Darkness - lighting unknown	Darkness - lights lit	Da
Clear Weather	1.16	1.18	
Fog or mist	1.10	1.14	
Heavy Raining	1.18	1.16	
Heavy Snowing	1.00	1.13	
Other	1.23	1.13	
Raining	1.11	1.15	
Snowing	1.08	1.09	
Windy	1.24	1.19	

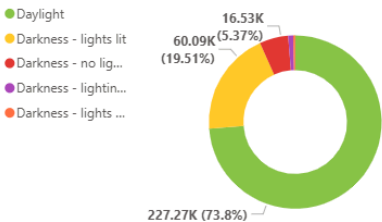


Environmental Factors & Infrastructure Impact

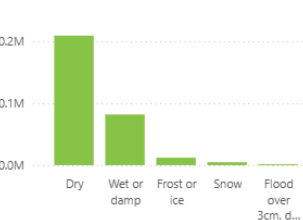
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Count of Accident_Index by Light_Conditions



Count of Accident_Severity by Road_Surface_Conditions



Weather Conditions Matrix by Lighting conditions

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Clear Weather	1.16	1.18	
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Other	1.23	1.13	
Raining	1.11	1.15	
Snowing	1.08	1.09	
Windy	1.24	1.19	

- Select Vehicle Ty...
☐ Agricultural vehicle
☐ Bus
☐ Car
☐ Heavy Goods Van
☐ Heavy Motorcycle
☐ High-Power Mot...
☐ Light Goods Van
☐ Light Motorcycle
☐ Medium Goods V...
☐ Minibus (8 - 16 p...
☐ Other vehicle
☐ Pedal cycle
☐ Ridden horse
☐ Standard Motorc...
☐ Taxi

- Weather Condition
☐ Clear Weather
☐ Fog or mist
☐ Heavy Raining
☐ Heavy Snowing
☐ Other
☐ Raining
☐ Snowing
☐ Windy

Select Month
All

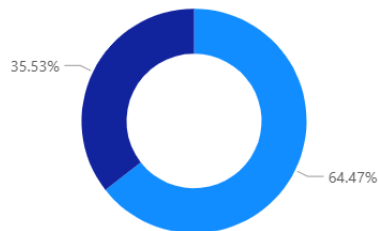
Day
All

Year
☐ 2021
☐ 2022

Geographical Analysis

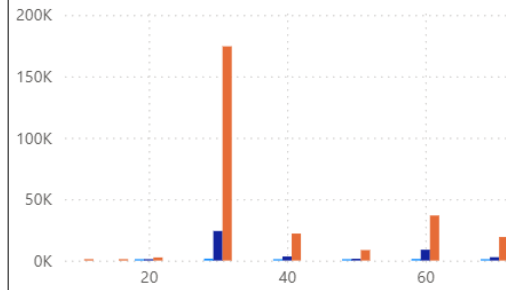
Total Accidents by Urban_or_Rural_Area

Urban Rural



Total Accidents by Speed_limit and Accident_Severity

Fatal Serious Slight



Geographic Risk Hotspots

Fatal Serious Slight



Geographic Risk Hotspot & Location Analysis

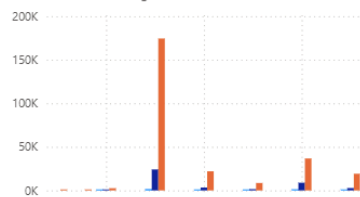
Total Accidents by Urban_or_Rural_Area

Urban Rural



Total Accidents by Speed_limit and Accident_Severity

Fatal Serious Slight



Geographic Risk Hotspots

Fatal Serious Slight



Select Vehicle Type

- ☐ Agricultural vehicle
- ☐ Bus
- ☐ Car
- ☐ Heavy Goods Van
- ☐ Heavy Motorcycle
- ☐ High-Power Moto...
- ☐ Light Goods Van
- ☐ Light Motorcycle
- ☐ Medium Goods Van
- ☐ Minibus (8 - 16 pa...
- ☐ Other vehicle
- ☐ Pedal cycle
- ☐ Ridden horse
- ☐ Standard Motorcy...
- ☐ Taxi

Select Month
All

Year
☐ 2021
☐ 2022

Day
All

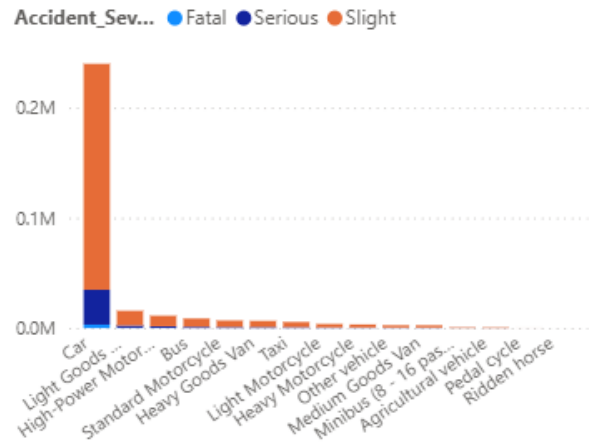
Weather Condition
☐ Clear Weather
☐ Fog or mist
☐ Heavy Raining
☐ Heavy Snowing
☐ Other
☐ Raining
☐ Snowing
☐ Windy

Vehicle & Casualty Analysis

Count of Accident_Index by Vehicle_Type

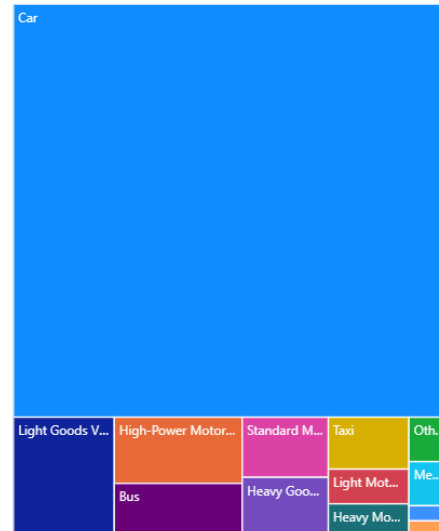


Count of Accident_Index by Vehicle_Type and Accident_Severity

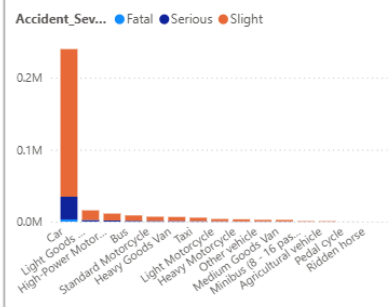


Vehicle Types & Casualty Analysis

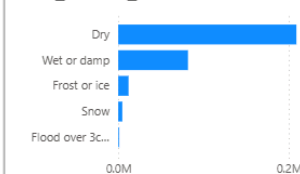
Count of Accident_Index by Vehicle_Type



Count of Accident_Index by Vehicle_Type and Accident_Severity



Count of Accident_Severity by Road_Surface_Conditions



Select Vehicle Type

- ☐ Agricultural vehicle
- ☐ Bus
- ☐ Car
- ☐ Heavy Goods Van
- ☐ Heavy Motorcycle
- ☐ High-Power Motorcycle
- ☐ Light Goods Van
- ☐ Light Motorcycle
- ☐ Medium Goods Van
- ☐ Minibus (8 - 16 passengers)
- ☐ Other vehicle
- ☐ Pedal cycle
- ☐ Ridden horse
- ☐ Standard Motorcycle
- ☐ Taxi

Select Month

All

Year

- ☐ 2021
- ☐ 2022

Day

All

Weather Condition

- ☐ Clear Weather
- ☐ Fog or mist
- ☐ Heavy Raining
- ☐ Heavy Snowing
- ☐ Other
- ☐ Raining
- ☐ Snowing
- ☐ Windy

Technical Implementation & Dashboard Features

Interactive Capabilities

- **Cross-page filtering:** Maintain filter context across all dashboard pages
- **Drill-down functionality:** Enable detailed analysis from summary views
- **Dynamic visualizations:** Real-time updates based on user selections
- **Export capabilities:** Support for data extraction and reporting

Performance Optimization

- **Efficient data modeling:** Optimized relationships and calculated measures
- **Responsive design:** Consistent performance across devices
- **Memory management:** Efficient handling of large datasets
- **Query optimization:** Fast refresh and interactive response times

Methodology & Analytical Approach

Exploratory Data Analysis Process

1. **Descriptive Statistics:** Understanding data distribution and characteristics
2. **Correlation Analysis:** Identifying relationships between variables
3. **Temporal Decomposition:** Extracting seasonal and trend components
4. **Spatial Analysis:** Geographic clustering and hotspot identification
5. **Risk Assessment:** Severity-weighted impact analysis

Visualization Strategy

- **Progressive disclosure:** From overview to detailed analysis
- **Visual hierarchy:** Emphasizing critical insights through design
- **Color psychology:** Intuitive severity and risk communication
- **Accessibility compliance:** Ensuring inclusive design principles

Strategic Recommendations for Infrastructure Planning

Immediate Priority Actions

1. **Junction Safety Enhancement:** Upgrade uncontrolled intersections with automated traffic systems.
2. **Lighting Infrastructure:** Improve illumination in high-accident dark zones.
3. **Peak Hour Management:** Implement traffic flow optimization during 15:00 hrs-18:00hrs. period
4. **Weather-Responsive Infrastructure:** Enhance road surface and drainage systems.

Medium-Term Infrastructure Development

1. **Urban Traffic Flow:** Redesign high-density accident zones with better traffic management.
2. **Speed Management:** Implement intelligent speed control systems in high-risk areas.
3. **Vulnerable Road User Protection:** Dedicated lanes and crossing facilities.
4. **Data Collection Enhancement:** Improve accident reporting and data quality systems.

Long-Term Policy Implications

1. **Smart Infrastructure:** IoT-enabled traffic management and real-time hazard detection.
2. **Predictive Analytics:** Use historical patterns for proactive infrastructure planning.
3. **Multi-modal Transportation:** Reduce private vehicle dependency in urban areas.
4. **Climate-Resilient Infrastructure:** Prepare for increasing weather variability impacts.

Data Quality & Limitations

Strengths

- **Comprehensive coverage:** Multiple dimensions of accident analysis
- **Temporal depth:** Sufficient historical data for trend analysis
- **Geographic scope:** Wide area coverage enabling spatial analysis
- **Multi-attribute analysis:** Rich dataset supporting complex correlations

Limitations & Future Enhancements

- **Data quality gaps:** 31.84% data quality issues requiring attention
- **Missing categories:** Some junction control and vehicle type classifications
- **Temporal granularity:** Potential for more detailed time-based analysis
- **External factors:** Integration with traffic volume and infrastructure investment data

Conclusion

This dashboard provides a robust foundation for evidence-based infrastructure planning and road safety policy development. The multi-dimensional analysis reveals clear patterns that can guide targeted interventions, resource allocation, and strategic planning decisions.

The interactive nature of the dashboard enables stakeholders to explore data from multiple perspectives, supporting both operational decision-making and strategic planning processes. The identified patterns in temporal distribution, environmental factors, and geographic concentration provide actionable insights for immediate safety improvements and long-term infrastructure development.

Key Success Metrics

- **Comprehensive KPI coverage:** All required metrics successfully implemented
- **User experience:** Intuitive navigation and interactive capabilities
- **Analytical depth:** Multi-layered insights supporting various stakeholder needs
- **Visual communication:** Clear and impactful presentation of complex data

This analysis demonstrates the potential for data-driven approaches to significantly improve road safety outcomes through targeted infrastructure investments and policy interventions.

To view the Entire Dashboard in detail, click here -

<https://drive.google.com/file/d/1wQZRdOUbj9JcXb4FoC8G3FrFn1XdJPYi/view?usp=sharing>

Created for: CiSTUP Data Analysis & Visualization Internship Selection Round-1

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